

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

December 2011

EE 305: Power Electronics and Drives -I

Date: 16.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) State and explain the requirements of ideal switching device. [05]
- (b) Draw symbol of Diac and GTO [02]
- Q - 2 (a) Explain gate characteristics of thyristor [04]
- (b) Explain construction and V-I characteristics of Silicon controlled rectifier. [06]
- (c) Explain briefly snubber circuit for thyristor [04]

OR

- Q - 2 (a) Explain construction and V-I characteristic of IGBT [04]
- (b) Explain construction and V-I characteristic of MOSFET [04]
- (c) Explain with neat diagram construction and V-I characteristics of TRIAC. [06]
- Q - 3 (a) Explain Resistance and Resistance-capacitance triggering circuit of thyristor. [07]
- (b) Explain briefly natural and forced commutation of Thyristor. [03]
- (c) Calculate the value of C_s shown in Fig. 1, so that thyristor is not turned on due to dv/dt . Also calculate the value of R_s so that maximum current through the thyristor does not exceed 20 A. dv/dt rating of thyristor is 200 V/ μ sec. [04]

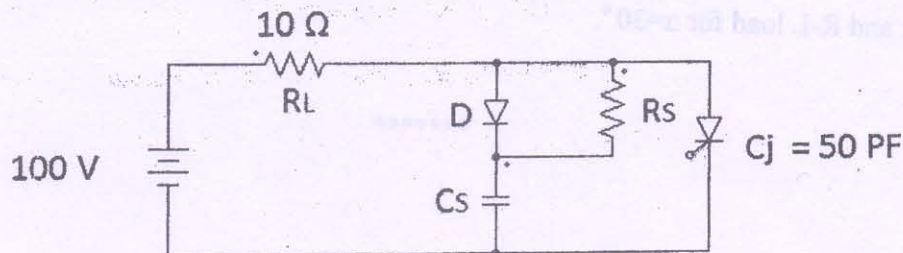


Fig. 1

OR

- Q - 3 (a) Explain in detail UJT & Diac based triggering circuit of thyristor. [07]

(b) Explain pulse transformer based driver circuit for SCR. [03]

(c) Explain series connection of Thyristors. [04]

SECTION – II

Q - 4 (a) Explain briefly different methods of speed control for dc motors. [07]

Q - 5 (a) Explain different control strategies used for obtaining variable output voltage from a dc chopper through appropriate waveforms. [07]

(b) Explain multi-quadrant operation of separately excited dc motor drive. [07]

OR

Q - 5 (a) Explain in detail Cuk Dc-Dc converter. [06]

(b) A step down chopper circuit working on Time ratio control principle is operating at a frequency of 1 KHz on a 220 V supply. If the load voltage is 165 V, calculate the T_{on} and T_{off} period of thyristor in each cycle. [04]

(c) A 220 V dc shunt motor runs at 1500 rpm at no load. The motor is fed through a type-A chopper. Find the speed of the motor when it draws 20 A current at $\alpha=0.5$ and $R_a = 1\Omega$. If the same motor is driven by an active load at 1710 rpm and input voltage is 250 V, which type of chopper will be suitable for speed control? Find the duty cycle if the armature current is limited to 20A. [04]

Q - 6 (a) Explain with neat waveforms working of 1- Φ , half wave phase controlled rectifier with R and R-L load for $\alpha=30^\circ$. [07]

(b) Explain operation of three phase semi-converter with Resistive load. [07]

OR

Q - 6 (a) Explain operation of three phase full wave fully controlled rectifier with Resistive load for $\alpha=30^\circ, 60^\circ$. [07]

(b) Explain with neat waveforms working of 1- Φ , full wave fully controlled rectifier with R and R-L load for $\alpha=30^\circ$. [07]

